

Programme	B. Sc. Computer Science				
Course Title	Programming using Python				
Type of Course	Minor				
Semester	III				
Academic Level	200-299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Have an understanding about algorithms and flowchart				
Course Summary	This course covers fundamentals of Python programming and teaches essential tools for data manipulation and analysis				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge	Evaluation Tools used
CO1	Understand the basic concepts of Python programming	U	C	Instructor- created exams / Quiz
CO2	Apply problem- solving skills using different control structures and loops	Ap	P	Coding Assignments/ Code reading and review
CO3	Implement simple Python programs to solve basic computational problems and GUI	Ap	P	Coding Assignments/ exams

	applications			
CO4	Analyze the various data structures and operations on it using Python	An	P	Instructor-created exams / Case studies
CO5	Apply modular programming using functions	Ap	C	Instructor- created exams / Quiz
CO6	Identify the necessary Python packages in the domain and create simple programs with it	U, Ap	C, P	Coding
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I		Problem solving strategies	12	15
	1	Problem analysis – formal definition of problem	1	
	2	Top- down design – breaking a problem into sub problems	2	
	3	Overview of the solution to the sub problems by writing step by step procedure (algorithm)	2	
	4	Representation of procedure by flowchart	1	
	5	Implementation of algorithms – use of procedures to achieve modularity.	2	

	6	Examples for algorithms and flow charts - At least ten problems Starting with non-numerical examples, and numeric problems like factorial, largest	4	
II	Introduction to Python		13	19
	7	Variables, Data types	2	
	8	Expressions and Statements, Evaluation of Expressions	2	
	9	Operators and Operands, Order of precedence, Boolean Expressions and logical operators, String Operations	2	
	10	Control statements, Conditional and alternative executions, Nested Conditionals, Recursion	2	
	11	Iteration - Multiple Assignment, While Statement	2	
	12	Tables, Two Dimensional Tables	2	
	13	Encapsulation and generalization, Local Variables	1	
III	Introduction to NumPy		12	18
	14	The Basics of NumPy Arrays, Computation on NumPy Arrays: Universal Functions	3	
	15	Aggregations: Min, Max, and Everything in Between	2	
	16	Computation on Arrays: Broadcasting, Comparisons, Masks, and Boolean Logic.	2	
	17	Fancy Indexing, Sorting Arrays	2	
	18	Structured Data: NumPy's Structured Arrays.	2	
	Functions		8	18
		Functions, Calling functions, Type conversion and coercion, composition of functions	2	

IV	19			
	20	Mathematical functions, User-defined Functions, Parameters and Arguments.	2	
	21	Strings and Lists – string traversal and comparison with List operations with Examples,	2	
	22	Tuples and dictionaries – Operations and Examples.	2	
	Hands-on Data Structures: Practical Applications, Case Study and Course Project		30	
Design programs from the concepts listed below. Select the topics and programs suited for your domain				
V	1	• Program to demonstrate basic data types in python • Program to demonstrate operators in python. • A cashier has currency notes of denominations 10, 50, and 100. If the amount to be withdrawn is input through the keyboard using input () function in hundreds, find the total number of currency notes of each denomination the cashier will have to give to the withdrawer. • Program to demonstrate list and tuple in python. • A library charges a fine for every book returned late. For first 5 days the fine is 50 paisa, for 6-10 days fine is one rupee and above 10 days fine is 5 rupees. If you return the book after 30 days your membership will be cancelled. • Write a program to accept the number of days the member is late to return the		

		book and display the fine or the appropriate message • Write a Program for checking whether the given number is an even number or not. • Write a Python program to print Fibonacci series. • Write function to compute gcd and lcm of two numbers. • Using a for loop, write a program that prints out the decimal equivalents of $1/2$, $1/3$, $1/4$. • Write a program to calculate overtime pay of 10 employees. Overtime is paid at the rate of • Rs.12.00 per hour for every hour worked above 40 hours. Assume that employee do not work for fractional part of an hour • Write a function reverse to reverse a list without using the reverse function.		
	2	Case study(Examples): ● Design a basic calculator application in Python that can perform addition, subtraction, multiplication, and division. ● Analysis of Antibiotic Resistance - Utilize publicly available datasets on antibiotic resistance in bacteria. Use NumPy to perform basic statistical analysis, such as calculating mean, median, and standard deviation of minimum inhibitory concentrations (MICs) for different antibioticss		

Reference Books:

1. Downey, A. et al., How to think like a Computer Scientist: Learning with Python, John Wiley, 2015
2. Lambert K. A., Fundamentals of Python - First Programs, Cengage Learning India, 2015
3. Sprankle, M., Problem Solving & Programming Concepts, Pearson India

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO 1	PO2	PO3	PO4	PO5	PO6
CO 1	-	1	2	3	2	2						
CO 2	-	1	2	2	-	-						
CO 3	-	-	2	3	2	2						
CO 4	-	-	2	2	3	3						
CO 5	-	-	3	3	3	3						
CO 6	-	-	2	3	3	3						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)

- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓	✓	✓	✓
CO 3	✓		✓	✓
CO 4	✓	✓	✓	✓
CO 5	✓			✓
CO 6	✓			✓